Week 12 Assignment

Mixed-effects models I

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 13's lecture (Monday). See the syllabus for the late policy.

Using this week's tutorial's simulated `field_data` (or re-simulate it with a different `set.seed()`):

1. Fit the naive model `lm(response ~ treatment)` and the mixed model `lmer(response ~ treatment + (1 | site))`.
2. Compare the standard error on the `treatment` coefficient between the two models. Which is larger, and why does that make sense?
3. Re-run the simulation with `n_sites <- 4` instead of 8. Refit the mixed model.

In 2–3 sentences: does anything about the model's output make you less confident in the random-effect estimate with so few sites?